

On Hopeful Monsters and the Origin of Language and Humanity: Challenging the Argument for a Founder Effect in Evolutionary Linguistics

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Abstract

For most scholars evolution by Darwinian principles of gradual accumulation of minor adaptive improvements is accepted as the standard explanation for the observable diversity of life forms. That said, the explanation of the notable gap between human cognitive and linguistic abilities and those of our evolutionary predecessors by evoking Darwinian gradualism has been unconvincing to some. An alternative explanation involving a founder effect, initiated by a hopeful monster, produced by a single genetic accident with transforming effects has been proposed. This saltationist argument has been particularly influential among students of evolutionary linguistics within the generative/ biolinguistic perspective in attempts to explain the unparalleled human capacities for language.

That said, the hypothesis proposed by the biolinguistic formalism have been challenged by empirical demonstrations which show divergence from the formal stipulations and demonstrate that there was never a single evolutionary event transforming pre-human into human species, but series of small and cumulative changes, resulting from multiple founder effects in the course of human evolution during Pleistocene. Moreover, human speciation is found to be a result of a combination of a number of evolutionary processes. A role for hybridization as a potential source of variation, complemented by genetic drift, found to have been major factors in the formation of the human phenotype, predominantly with neutral selective consequences, in interplay with selection offers a novel, integrative approach, presenting a significant challenge for the founder effect as a credible explanation for human evolution.

Such challenge is extended to the bio-linguistic argument as explanation for the human capacity for language, i.e. both Language of Thought and Universal grammar.

The argument for a founder effect in diversification of languages from a “mother tongue”, based on assumed parallels between diversification of populations and languages, is challenged by demonstrations of divergence in migration patterns, population size and historical and cultural idiosyncrasies and favours an explanation for linguistic diversity resulting from self-organization.

Keywords: language evolution, Universal Grammar(UG), language faculty(LF), Modern Synthesis(MS), Extended Evolutionary Synthesis(EES), founder effect,

1. Introduction: Darwinian gradualism and its challenges in modern evolutionary thought

In the theory of evolution in all its interpretations the evolutionary process is defined as a continuum of successive stages of heritable variation produced by random genetic mutations, subjected to environmental selection, each judged by their functionality in certain environmental circumstances. And although the gradualist argument is widely accepted, some question the explanation of biological novelties with accumulation of small successive adjustments, especially where examples of evolutionary continuity are unavailable. A plausible

explanation for the apparent evolutionary discontinuity is that the sources of biological innovation are different in different circumstances, e.g. gradual, slow adaptive changes occur in large populations, while relatively rapid, abrupt jumps with peaks in activity, which disturb the usual stability, happen in small, isolated populations. In this sense Joseph and Janda (2003) find the debate on gradualism vs. saltationism irrelevant as each of these alternatives explain the same facts from different viewpoints, involving different time-scales, i.e. geological (where time is measured in millennia) and biological (which measures changes in a few generations). The argument is that on a geological scale gradual changes, which are normally small and cumulative, are undetectable and thus, are registered as stasis. Changes are registered only when the difference of before and after is significant enough to be detectable, which explains the apparent gaps in the material remains they have left behind in the paleontological record. Small events of little significance which bring quantitative alterations, each building on the next, sooner or later result in qualitative transformations. Thus, small changes are only noticed when they accumulate and produce evolutionary Rubicons as quantity transitions into quality. Similarly, as admitted by Thieben (2009), the world of biological forms rarely lends itself to clear and sharp theoretical distinctions and absolutes are difficult to pinpoint, e.g. the clear differentiation of cumulative evolutionary innovation, which implies gradual modification of known antecedents, and an innovation produced by a single event, is difficult to establish. Moreover, the matter is further complicated by the fact that species differ in the length of life cycles, e.g. time passes differently for plants, bacteria and mammals, which makes a uniform criterium for rapid change difficult to establish and which injects confusion in the saltationist argument.

A theoretical alternative to Darwinian gradualist perspective as the only explanation for life diversity is first articulated by R. Goldsmith who in the 1940s who, although agreeing that gradualism is the best explanation for phyletic diversity, points at some cases, isolated and rare, where macro-mutations produce evolutionary novelties for which a gradualistic explanation is not plausible. In such cases a single individual specimen, a genetic carrier of a novel, unprecedented property, i.e. one for which antecedents are not traceable, is produced. And although in the majority of cases such genetic discontinuities fail to procreate, which precludes their influence in evolution, prompting the label “hopeless monsters”, in rare occasions such genetic discontinuities manage to survive and engender new species, arising abruptly within a short evolutionary window without antecedents, prompting the label “hopeful monster”. Consistent with this line of reasoning is the argument which explains evolutionary novelties with alternation between sudden bursts of biological creativity and mass extinctions, articulated in the hypothesis of Punctuated Equilibria (Gould, Eldridge, 1972). The empirical support for either is limited to certain species and certain periods.

In attempt to incorporate evolutionary accidents into the evolutionary theory Gould and Vrba (1982) have enriched the anti-adaptationist argument with the concept of exaptation, a special case of adaptation where under certain circumstances an organ or a system initially functionless, a spandrel, undergoes a process of exaptation, i.e. undergoes evolution as adaptation for a function. Exaptation is also evoked when a biological entity, evolved for a certain function, receives a secondary function, different from the original one in response to sudden environmental change. In this case adaptation to the new function is constrained by its antecedents. Necessarily the new function must be maximally similar to the original one for the exaptation to be successful. This is one way evolution looks for shortcuts by recycling what is

already available in its attempt to respond to rapid environmental changes. The anti-adaptationist argument has had little success in convincing the majority of evolutionary biologists.

2. Evolution of the evolutionary theory

2.1. The Modern Synthesis and genetic determinism

The discovery of the gene triggered reevaluation and extension of the original Darwinian theory resulting in the Modern Synthesis (MS) as a new and improved understanding of evolution. The Mendelian model of evolution bears the name of Mendel who in the 19th century designed a mathematical model for describing heredity. Mendel's theory understands the organism in terms of an abstract model as a combination of discrete elements and describes heredity as “inheritance by which stable non-decomposable characters, controlled entirely or overwhelmingly by a single genetic locus, are transmitted over many generations” (in www.medical-dictionary.thefreedictionary.com/mendelian+inheritance). It is influenced by classical physics and studies evolution in terms of recombination of stable immutable entities which act independent of context. Mendelian use of mathematical concepts is consistent with the Turing's theory of computation where a finite and predetermined set of abstract symbols automatically combine according to a finite number of equally predetermined rules producing an infinite number of combinations. In this sense evolution is defined as a computation process where a finite number of basic discrete units of DNA molecules, which form genes, combine and form the diversity of life forms and heredity is understood in terms of digitalization.

2.2. Population genetics and the hypothesis of a founder effect in evolution

Population genetics has emerged as a branch of MS which studies the genetic reservoir in populations and its evolutionary role as a source of variation on which natural selection operates. The study of population genetics is instrumental for evolutionary theory as evolution is defined in terms of change in the genetic makeup of populations. It introduces concepts, *genetic hitchhiking*, *genetic drift*, *founder principle*, *gene flow*. Genetic hitchhiking happens when the frequency of an individual gene changes due to its pertaining to a chromosome, i.e. the distribution of a single gene depends on the distribution of the chromosome of which it is a part. Gene flow is defined as migration of genes from one population to another as a result of migration of individuals. It assumes that evolution occurs in populations and explains the evolution of species of social animals, e.g. bees, ants, dogs, penguins, humans, where the evolutionary fate of the individual is dependent and coordinated with that of conspecifics. Genetic drift is a process of reduction of genetic diversity within a small reproductively isolated population by change in the frequency of an allele, reducing variation in a population. Mayr (2001) introduces “the founder principle” to define cases when a new genetically different population or a new species evolves from a chance mutation in a single individual, rapidly proliferating within a small reproductively isolated population. Population genetics has important contributions to understanding evolution by designing mathematical models for calculating probability of gene distribution in populations.

Crucially, population genetics regards adaptation as one of various factors in the evolution of small isolated populations as change in gene frequency is in addition attributed to genetic drift, gene flow, often producing and perpetuating maladaptations, as illustration of counterexamples for the relevance of adaptation, and its role in determining the directionality of evolution. So, evolution is viewed here as partly guided by factors of population dynamics, i.e. insulated from environmental interference, as opposed to the classical view of evolution from without, i.e. imposed by the environment. (Lynch 2007).

2.2.1. The hypothesis of a” hopeful monster”

“ In evolutionary biology the concept of the founder effect refers to “ the establishment of a new population by a few original founders(in the extreme case by a single fertilized female) which carry on only a small fraction of the total genetic variation of the parental population”. The resulting population thereby instantly becomes genetically different from the parental population.” (Jared Diamond, Jerome Rotter, Observing the founder effect in human evolution, Nature, vol.329,1987, p.105)

A founder effect is accompanied by the loss of genetic variation resulting from geographic and reproductive isolation from a parental population most often during colonization of new territories. It is a significant factor in the evolution of populations. The argument for evolutionary innovation caused by a genetic macro-mutation producing an evolutionary novelty in a single specimen has been unconvincing for the majority in the field of evolutionary biology, given that a genetic incompatibility would not lead to successful reproduction and , in addition, a genetic innovation does not necessarily, even rarely, ensure adaptability, therefore highly reducing or completely eliminating the mutant's chances to become a progenitor of a new species.

And given that the hypothesis of the “ hopeful monster” is an extension of genocentrism, i.e. MS, its limitations translate in limitations of the MS(see Theiben 2006, 2009 for discussion).

2.2. Extended Evolutionary Synthesis and its conceptual innovation

The latest empirical findings demonstrate that in biology pleiotropy (a single gene influences multiple, often unrelated phenotypic traits) is the rule, not the exception given that a change in a single gene most often than not has no significant effect on a trait, suggesting that the genome is inherited as a whole. This explains why despite the frequency of mutations, the genome remains remarkably stable. Moreover, all cells in an organism inherit the same genetic information and yet they develop into specialized tissues and organs, very different in structure and functions, e.g. skin, liver, etc. and follow developmental schedules, specific for the type of cells. Individuals with almost identical genetic material (twins) grow up as different individuals, and non-human primate species which share 98% of genome with humans display vastly different behaviours and physical appearances, suggesting that inheritance cannot be reduced to the genome. Significantly, it is now known that the genome is made of different types of genetic material, a good part of which has no role in the formation of the biological body and no exposure to selection, while other components of the cell are inherited and participate in the building of the phenotype on par with the genes.

This prompted the argument for revision of the MS and conceptual enrichment of the

evolutionary theory, i.e. EES, which introduced new concepts, e.g. extragenetic inheritance, epigenesis, niche construction, reciprocal causation, gene-culture coevolution to label the multiplicity of factors and processes, epigenetics, development, learning, the environment, which interact and cooperate in a multidimensional evolutionary process in the formation of the organism by multiple sources of inheritance(Jablonka, Lamb, 2005 and elsewhere)

The EES emerges from the finding that the DNA molecule is only one component of the cell structure, as heritable information is contained beyond genes at cellular level where cell memory contains instructions, specifying the differential formation of organs and systems, i.e. the cell is inherited as a whole. The cell functions by cooperation of different types of molecules and multicellular organisms function by cooperation of multiple cell types. Most importantly, this reveals that the role of a single gene can only be understood through its integration and interaction with other mechanisms of inheritance. The same genetic material is “packaged” differently in different cell groups dependent on their function in the phenotype and interpreted differently during development. Cell memory includes information on patterns of gene actuation through regulatory genes which control the timing and location of turning on/off of gene functions during development as different DNA sequences are activated at different times and chromosome locations.

The term “*epigenesis*” was coined by Waddington in the 1940s and refers to processes of formation of the phenotype, through interaction of the genome with the environment at the molecular level. Epigenetic mechanisms “regulate genomic activity independent of DNA sequence”(Skinner, Nilsson 2021, p.3; for details on terminology of epigenesis see Jablonka 2017), therefore, genomic activity is conditioned on epigenetic processes. The environmental influence on genomic activity thorough epigenetic processes produces variation at the level of cells, organ tissues and morphology. Temperature, stress, nutrition, even the sun, etc. are environmental factors triggering epigenetic response, although epigenetic processes can be triggered by organism-internal and intracellular factors. The epigenetic routines are also referred to as “canalization” as they constrain and shape the developmental process to follow established pathways(Jablonka, Lamb, 2005 and elsewhere). The term “*canalization*” was introduced by Waddington to mean reliable preservation of developmental routines by insulating these from possible perturbations caused by environmental or other influences. Epigenetic mechanisms also evolve.

While in MS context, where the focus is exclusively on genetic determinism and the role of non-genetic inheritance is viewed as negligible, the proponents of EES argue that epigenetic mechanisms are active participants in the overall evolutionary process on par with genes (Jablonka, Lamb, 2005 and elsewhere). As per Skinner, Nilsson (2021) epigenetic changes are five times more frequent than genetic mutations, suggesting that epigenesis has crucial role in variation and adaptation, i.e. crucial for evolution.

The role of development in evolution is the focus of the developmental systems theory (DST)(West-Eberhard 2003; Griffiths, Hochman, 2015) in its approach to evolution and heredity, which emphasizes the role of extra-genetic factors as significant contributors to the formation of the phenotype and views genes as one among multiple ingredient in the process of phenotype formation. The unit of selection is the life cycle and its replication, i.e. the biological organism at its developmental stage is the unit of selection as it determines the viability of the adult organism. DST attributes the formation of the biological form to phenotypic plasticity where the environment influences the pathway of the organismal development, which in turn

alters the genetic makeup, i.e. the causality is reversed from phenotype to genome through development, i.e. genetic mutations are the result, not the cause. The theoretically established dichotomy between genes and environment is rejected as misleading. Similarly, West-Eberhard (2003) argues that phenotypic novelties are generated with the mutual accommodation of genotypic and environmental factors interacting during development. Consequently, diversification of species by epigenetic variation can explain evolutionary puzzles inexplicable by genetic evolution.

2.2.1. Evolution as adaptation by learning

Biological inheritance systems of genes and developmental mechanisms are not the only vehicles for evolutionary change, as in many species adaptation is accomplished by adopting new behaviours either by individual experience through trial and error, i.e. individual learning, or/and by observation and imitation of behaviour of conspecifics, i.e. social learning. Moreover, new behaviours are selectively perpetuated by the next generation only on the basis of their superior survival benefits to alternatives, i.e. learned behaviours are subjected to evolutionary processes. And learning, if adaptive, influences the trajectory of evolution. Moreover, in species with advanced learning abilities, e.g. some primates and most notably, humans, learning has crucial role in adaptation by formation of systems of culture, where products of culture, e.g. material objects and behavioural practices, become integral components of the environment.

In addition, species and environments are not independent entities. For one thing, animals are not passive experiencers of environmental pressures, they chose the environment in which they live and adapt to. Not only the environment influences life forms, life forms influence the environment through niche construction. All species, from insects to humans, alter the environment in which they function through their behaviour. The environment and life forms form an integrated complex, the components of which evolve in tandem by a process of co-evolution, a process of reciprocal evolutionary change which occurs in entities (species, groups of species) as a result of their interaction and interdependence.

“...Taken together the relations of genes, organisms and environments are reciprocal relations in which all three elements are both causes and effects. Genes and environment are both causes of organisms, which are, in turn, causes of environments...” (Lewontin, 2002 p. 100).

Thus, evolution is a multidimensional process with multiple components interacting in different contexts and “Something like evolution can occur in each dimension... we are a messy complex. And it is the complex that evolves” (Lamb, Jablonka, 2005, p. 231).

2.3. The founder effect, an evolving concept

In attempt to circumvent the limitations of MS later interpretations of the hopeful monster hypothesis argue for a crucial role of regulatory genes, and their developmental effects on the appearance of evolutionary novelties, i.e. the sudden appearance of genetic peculiarities is attributed to developmental mechanisms by which the timing and place of gene's functioning is controlled. In this context the appearance of groundbreaking novelties is explained with sudden

alterations of developmental routines, culminating in the appearance of significant morphological deviations from parental organisms. Thus, hopeful monsters are individuals with novel morphological constitution, produced by extragenetic mechanisms, products of an event, justifying the characterization of saltation (see Theiben, 2006, 2009).

Importantly, saltation in the appearance of evolutionary novelties is effectively demonstrated in plants, e.g. in diversification of some flower species, and rarely in animals, where the most convincing example is the appearance of turtle body plan, for which so far no antecedents are found in the record, eliminating a gradualist explanation, and strongly suggesting a possibility for saltational origin by a hopeful monster (Theiben 2009).

The above mentioned isolated examples of putative phyletic revolutions occurring predominantly in the morphology of simpler species, whose developmental cycles do not involve diverse developmental mechanisms and processes, I imagine, stand a good chance to be a convincing enough claim for the saltationist account. That said, such claims are much less convincing when it comes to explaining discontinuities in cognition and especially in human cognition and language, which is why the majority of students of language evolution remain skeptical as to a saltationist explanation, to be discussed further in later segments.

3. The founder effect in human speciation and its empirical challenges

The hypothesis for a “hopeful monster” as a highly unusual case of founder effect is evoked by some as explanation for human speciation, demonstrated by the unprecedented cognitive and linguistic abilities of humans. A founder effect initiating the human speciation by a single founder (perhaps a single female) as a carrier of a ground-breaking mutation within a small population of spatially isolated homo population is estimated to have occurred in the last 100,000 years. That said, a number of genetic studies, which trace the evolutionary history of the human species during Pleistocene (2.8my-11,000 ya), challenge the hypothesis on several major points.

a. The likelihood that a founder effect could result in speciation is very small.

Various authors find that the generation of new species necessitates genetic drift to act in perpetuity for a long period over a number of generations, i.e. a speciation event in a single generation is unlikely. As per Diamond, Rotter (1987) all along human evolution the human genome is sculpted by multiple founder effects in small populations of about 500 individuals, suggesting inbreeding, as in polygamous families. This opens a possibility for some deleterious mutations, e.g. proliferation of heritable genetic disorders, and as per Matute (2013) inbreeding commonly results in extinction. At the same time most mutations are benign, e.g. skin pigmentation, resistance to some diseases. For example the genes resistant to malaria in populations of sub-Saharan Africa result from adaptation to environment under the influence of the founder effect. The effects of reproductive isolation from founder effects dissipate after population expansion within 30 generations (Matute, *ibid.*). Importantly, it is suggested that reproductive isolation could result in novel phenotypic traits and behaviours, adaptive to a new environment by exposing hidden phenotypic plasticity, not by genetic effects, which usually result in decrease of fitness. So, “reproductive isolation is more likely to evolve through gradual evolution than through genetic revolutions” (Matute *ibid.*). In sum, so far the theoretical

framing of human speciation by founder effect is not supported by empirical findings.

b. No evidence of a single event of genetic transformation

As per Hawks et al. (2000, see also Pollen et al. 2023) the data does not support the hypothesis of a recent major genetic event of speciation resulting from a genetic bottleneck, which would have left traces at various genetic loci and significant phenotypical changes in anatomy, physiology, etc. Instead, anatomical and archeological evidence support the argument that “there may have been a number of severe population size bottlenecks on the lineage leading to living humans...because of the many speciation events that must have occurred” during the divergence from ape species in the last 6 my of Pleistocene (Hawks et al. p.3.) This suggests that wild demographic fluctuations are nothing new or unique in the history of human speciation.

At the same time perhaps the most significant speciation event in human evolution, due to isolation and founder effect, dated at 2 mya is argued to have produced *Homo Erectus*, or *Homo Ergaster* as a genetic diversion from *Australopithecus*. Phenotypic demonstration of such genetic changes are noticeable not only in increased cranial size, suggesting brain enlargement, but also in numerous aspects of human anatomy and physiology, e.g. dentition, enlargement of body size, leg length, vision, respiratory system, etc. Significant changes in energy demands associated with such biological changes are coordinated with increased meat consumption, socialization for food sharing, etc. Behavioural changes, e.g. manufacturing of tools of some complexity at about 1.5 my are exemplified by the Acheulian industry, first to emerge in East Africa. So, a significant and sudden genetic transformation within a small population of *Australopithecus* paves the way to *Homo sapiens* with intermediate speciation events producing multiple new species. Thus, a single genetic event of divergence from apes, initiating the *Homo* branch, much earlier in human evolution, followed by a number of additional, perhaps equally transformative genetic events during a protracted period have shaped the human genomic specifics, provoking changes in physiology, cognition, behaviour, brain architecture, immune response, etc. suggesting gradual transitions with cumulative effects.

On the other hand, an alternative hypothesis rejecting the possibility of a population bottleneck would imply a continuous genetic progression from several million years to the present, leaving genetic traces of ancestral species older than 2 my, expected to be detectable in the human genome by much broader genetic diversity, which is not empirically demonstrated.

Interestingly, as per Pollen et al. (2023) a large part of genetic changes are detected in non-protein coding parts of the genome, suggesting that these had no exposure to selection and involve structural changes in chromosomes, e.g. insertion, deletion, fusion etc. implicating 3% of the genome.

c . Population size at bottlenecks

The dominant view postulating a recent and significant reduction of population size of about a thousand to a few thousands from a bottleneck, followed by a “massive Late Pleistocene population expansion” is challenged by genetic and mitochondrial evidence which suggests “no recent significant bottleneck in human prehistory” (Hawks et al. 2000, p. 10-11, also Ayala et al. 1995). Instead, a long period of persistent small population size, i.e. a “long-necked bottle”

of numerous populations spread on vast territories at low density of a few thousand individuals is a better interpretation of empirical findings. e.g. 22 adult couples inhabiting the current territory of France and 35 couples living on the territory of today's Germany following the colonization of Erectus out of Africa. And given the evidence of sharp climatic fluctuations during Pleistocene and resulting abrupt changes in habitats, subdued population growth and even occasional depopulation is to be expected. Thus, no significant population expansion before 10,000ya is currently supported. In sum, genetic and anthropological findings converge in supporting the hypothesis that there was a population bottleneck at about 2 mya, which produced significant genetic divergence from Australopithecus with profound anatomical, cognitive, behavioural consequences, followed by series of minor genetic transformations, resulted in the rise of the human phenotype.

d. The human phenotype, a product of “ admixture” of contributions from various hominid genotypes.

Genetic contributions from great apes e.g. chimpanzee, bonobo, gorilla, various species of orangutang are found in the human genome. Hominid genome contributions are significant and vary by populations, e.g. genetic contributions of Denisovans are most significant in Melanesian and Australian aboriginal populations, i.e. up to 5%. The Neanderthal genetic ancestry is about 2% in all non-Africans as evidence for interbreeding. Many genetic diversions from the Neanderthal genome are changes in gene regulation involving transcription processes in different genomic locations(Pollen et al. 2023, p. 691) altering the developmental schedules with consequences for the human phenotype. Especially significant are changes resulting in divergence in brain development and maturation schedules. Epigenetic changes are found to influence the gene expression of several hundred human genes. Significantly, “Cumulatively, it is estimated that at least 20-40% of Neanderthal DNA survives in human populations around the world.” where“ Neanderthal alleles are associated with skin and hair colour, immune response... skull shape, bone morphology, sleep patterns, mood disorders.” (ibid.Pollen et al. 2023, p.689).

In sum, both earlier and more recent studies suggest that: **a.** There was never a single evolutionary event transforming pre-human into human species, but series of small and cumulative changes, resulting from multiple founder effects in the course of human evolution during Pleistocene. **b.** The argument for a founder effect, i.e., highly reduced population, even a single female, spatially and thus, reproductively isolated from ancestral homo populations, as a genetic founder of the sapient species, is inconsistent with recent findings by population genetics which fails to confirm such drastic reduction of primate populations. On the contrary, population numbers of 1000-10,000 range and even larger has been the norm throughout the Pleistocene period. **c.** The argument for the initiation of a new type of species resulting from a complete genetic cutoff from ancestral populations is challenged by findings that the human genome shows small cumulative genetic contributions from a number of Homo populations, to the genetic configuration of Sapiens. But most significantly, despite the focus on genetic evolution, perhaps the most significant changes are not to be found in the genome, but in epigenetic, developmental and learning contributions, which complement meaningfully to the physiological, cognitive and behavioural peculiarities of humans.

3. 1. Human speciation, a messy combination of evolutionary processes

The quest for human origins until recently was focused on understanding the role of the genome by tracing the succession of genetic mutations which lead from Australopithecus to Sapiens, where genetic changes were explicable from adaptationist perspective with functionality, while only secondary attention was given to phenotypic diversification. As an alternative to the adaptationist account a population-genetic perspective argues for a single genetic event with transformational effect, or series of genetic transformations, insulated from environmental factors, with smaller effects in isolated populations, cumulatively produce the human phenotype.

That said, the story of human origins is proving to be much more complex and dynamic, where diverse evolutionary forces have interacted at different times and to different degrees in different locations as the human body and mind were shaped. Schroeder and Ackermann (2023) offer a new and improved with debt and detail hypothesis of human evolution, which diverges in many ways from currently dominating hypotheses. For example, evidence is shown for the role of selection in brain enlargement, but not for body enlargement during hominid evolution. In this sense it is found that “ substantial number of traits across the skeleton in humans have evolved neutrally (neutral to adaptation, emphasis added)” (ibid. p.6). Moreover, it is now known that genetic drift and interbreeding, processes neutral to adaptation may have contributed to the phenotypic diversity within populations . The neutral theory stipulates that multiple, equally well adapted phenotypes, are produced by random genetic drift. Significantly, the theory also states that most changes occur at molecular level in non-coding parts of DNA, i.e. irrelevant or neutral to adaptation.

In addition, hybridization, i.e. interbreeding between two species and/or subspecies, traditionally viewed as resulting in reduced viability and even sterility, as per recent findings, summarized in Schroeder, Ackermann (ibid. 3.3) has positively impacted human evolution. For example interbreeding with Neanderthals and Denisovans is now established, although it is difficult to determine if and to what extent genetic contributions have resulted in increased phenotypic adaptability.

In sum, empirical findings furnish the understanding of human speciation as a result of a combination of a number of evolutionary processes. A role for hybridization as a potential source of variation, and genetic drift, assumed to have been a major factor in the formation of the phenotype, predominantly with neutral selective consequences, in interplay with environmental selection, offers a novel, integrative approach, with potential for a better understanding evolutionary path our ancestors have taken towards becoming human. And given that linguistic abilities are regularly regarded as the hallmark of human speciation, from the perspective of evolutionary linguistics “ the evolution of language is ...far less exceptional by biological standards, (where)...there were potentially many mutations involved that had small to moderate fitness effects that all contributed to the ability for language ” (de Boer et all. 2020, p.3)

3.2. Rethinking the evolution of cognition: Evolutionary Psychology vs. New Thinking

Theoretical perspectives on evolution have influenced hypotheses and research on the evolution of human cognition as integration of evolutionary principles in cognitive science. Evolutionary

Psychology, a paradigm which has dominated the field of cognitive science during the last few decades(Tooby, Cosmides, 2015) understands the human brain as a conglomerate of multiple, computationally distinct and spatially isolated processing subsystems, or modules, each controlling a single aspect of human behaviour. From evolutionary perspective the hypothesized “ massive modularity” is explained as shaped by adaptation within the Erectus brain in response to the cognitive and behavioural demands of our ancestors during Pliocene, i.e. since the last 2 my. In this sense human cognition has evolved for solving problems of the remote past, e.g. avoiding predators, cheater detection, food preferences, mate preferences, dealing with the challenges of communal living, e.g. form coalitions, resolve conflicts, etc. And given that human cognitive abilities, supported by brain architecture and processing mechanisms, has evolved by adapting to past challenges, it is said to be ill equipped for solving problems of the present , since our current environment, natural and social, has changed dramatically, especially in the last 11,000 years especially with the emergence of the first civilizations. Evolutionary Psychology adopts Mendelian genetics and MS interpretation of evolution as it aims to explain the formation of cognitive modularity with a cascade of genetic mutations within the ancestor brains, which within a short evolutionary time is argued to have produced the human brain. In other words, Evolutionary Psychologists hypothesize that “ new cognitive processes appeared suddenly and fully formed as a result of lucky genetic mutations “(Heyes, 2012, p.2093). Given their genetic underpinnings, cognitive modules are purported to develop uniformly and without environmental input and operate subconsciously and automatically when activated by environmental stimuli. Thus, the human mind is defined as a compilation of a number of structurally and spatially isolated processing mechanisms, each controlling a single behaviour, and a product of gene-based inheritance.

Recently an alternative vision of the human mind and its evolutionary origins has emerged, or a “new thinking” (Heyes, 2012) which challenges the very foundations of the paradigm of Evolutionary Psychology. To begin with, the mind is defined as emergent from initial state of domain-general cognitive computations, able to process a broad range of experiences, which, although displaying evolutionary continuity with pre-human ancestors, are at the same time endowed with unique cognitive power and flexibility in humans. Thus, what is innately available at birth is a powerful general purpose computing entity. From this initial point developmental processes unfold incorporating experience and the co-evolutionary processes between the two produces domain-specific processing mechanisms, i.e modularization of the adult mind is a product of interaction of nature and learning. In this sense a reasonable explanation for the evolution of language is articulated (Dor, Jablonka 2000 and elsewhere), as having originated as a set of conventions, initially learned by domain-general mechanisms, followed by a process of mutual adaptation of language and human cognition, resulting in the internalization, or genetic assimilation of some fundamental properties of language by an evolutionary process termed the Baldwin effect, i.e. by converting nurture into nature.

4. The founder effect as explanation for the evolution of language

The postulation of a founder effect is a crucial factor in the attempt by the generative/ biolinguistic perspective to offer a credible explanation for human linguistic abilities. Central in this endeavour is a formal model for a Language Faculty(LF) which, despite being convincing as a formalism, has encountered difficulties in demonstrating empirically the factual existence

of LF, i.e. its specific location and functions in the human brain, and for which credible evolutionary explanation is also problematic. Earlier formalisms within the generative respective postulate a LF with computational specifications difficult to translate in bio-cognitive instantiation, especially the postulation of its instantaneous appearance fully formed without any precursors, i.e. in violation of standard evolutionary principles. An attempt to overcome this explanatory obstacle was to adopt the theoretical perspective of population genetics, especially the claim of proliferation of counter-adaptive and even maladaptive traits in small populations as a result of a founder effect initiated by a hopeful monster. In this context the LF was attributed to a genetic accident in a genius mutant, a carrier of a gene, affording unprecedented cognitive power, effectively far beyond the needs for biological survival, or as per Bickerton “more than nature needs”. Within the population genetics perspective such genetic accident in a small population of Homo species, although irrelevant to adaptation, or even counter-adaptive, is hypothesized to have proliferated and became the progenitor of the Sapient species (Bickerton 2014 and elsewhere), making the explanation for human linguistic abilities a challenge for the theory of evolution.

3. 1. The biolinguistic explanation for a Language Faculty(LF) and its challenges

In the biolinguistic argument the hypothetical LF is envisioned as a cognitive module designed to perform unlimited number of recursive computations on linguistic primitives, e.g. words, morphemes, producing complex sentences. The standard generative / biolinguistic perspective explains it as the phenotypic result of a one-time genetic event, an unspecified felicitous mutation, is articulated in detail in Chomsky (1995) and recently in Berwick, Chomsky (2016).

3.1.1. Language of Thought (LoT), the foundation of linguistic semantics

As per the generative/biolinguistic perspective the functioning of the LF is pre-conditioned on a cognitive antecedent, another significant cognitive transformation in the mind, the appearance of a language of Thought (LoT), a novel type of cognitive processing mechanism, performed by a cognitive module, designed to produce unlimited number of recursive computations on conceptual primitives, generating complex propositions. Its appearance is said to initiate the transformation of the ape cognition into human, hypothesizing a “Great Leap Forward”, or the birth of new cognitively distinct species, knowledge seeking Sapiens species. The unprecedented cognitive entity, a computing mechanism, is said to be capable of deriving all concepts ever conceivable by human minds from a small number of abstract conceptual atoms, organized in structured combinations under abstract rules (Fodor 1975, Fodor, Pylyshin 2015). The abstract rules, or “laws of thought” are said to be similar but not identical to the rules of logic. Crucially, LoT defines meaning as abstract, that is, independent of perceptual experience, innately predetermined, universal, and digital, features highly unusual for a bio-cognitive entity. Moreover, despite its cognitive sophistication, the LoT is assumed to have appeared in the human mind in its complete form at once, without precursors, an evolutionary Big Bang, diverging from fundamental evolutionary principles of continuity. Human prefrontal cortex is known to be the location of serial embedding of complex abstract structures, i.e. the LoT, used in a variety of behaviours, e.g. problem-solving, music, language, tool manufacturing etc. (Wang et al. 2019).

The LoT hypothesis is a purely formal conceptualization of the human mind. Attempts have been made by the evolutionary psychologists Tooby and Cosmides(1994 and elsewhere) to provide empirical evidence for compartmentalization of mind and for the LoT, as formally conceived. Nevertheless, in all life forms cognition must reflect experience because cognitive capacities exist and evolve only if relevant to survival. Defining human cognition as independent of perceptual reality and experience with it and in terms of modern artificial systems makes this argument implausible.

The hierarchical organization of human thought is said to be the evolutionary foundation for the LF, produced by the accidental association with a grammar-processing algorithm, envisioned to have appeared in the brain as a resolution of a brain-internal conflict between the increased anatomical volume of the brain and the spatial constraints of the skull. As a result, the grammatical concepts and computational principles were imposed on LoT, producing a highly complex computing entity for processing linguistic meanings and their configurations. Thus, in generative/biolinguistic context the formation of the LF is envisioned in terms of adding computational sophistication to human reasoning in terms of syntactic categories and highly abstract rules of their association.

A new interpretation of LoT by acknowledges the role of perception in cognition and attribute LoT-like properties to perception as the two systems are found to interface. Moreover, elements of LoT are found in pre-linguistic infants and non-human primates in departing from the original stipulation of LoT as uniquely human aspect of cognition, suggesting cognitive continuity. In addition, it is acknowledged that human cognition is more than symbol manipulation and incorporates a role for learning and experience(Quilty-Dunn, Porot, Mandelbaum 2022)

“ We believe in other vehicles of thought, associations, icons and much more.(*ibid.* in the Conclusion)

3 .1.2. Merge, the essence of LF: theoretical and empirical challenges

The computational operation Merge, which assembles individual conceptual primitives into linguistic structures of unbounded length and complexity, is the heart of the biolinguistic understanding of LF as its essential property. Within the minimalist approach it is said to exhibit maximum computational simplicity and efficiency, a stipulation which, by comparison of the minimalist version with earlier predecessors of the generative formalism, is certainly true.

That said, Martins, Boeckx (2019) contest the argument for computational simplicity of Merge, given that as per the Minimalist Program Merge comes in two versions: *external* Merge by which conceptual primitives are successively combined forming conceptual structures of ever increasing complexity, and *internal* Merge, by which additional complexity is injected by creating cross-boundary structural dependencies. Thus, Merge cannot be regarded as a simple operation as its application implies computational constraints on where and when new additions to the structure can and cannot be positioned. Moreover, the performance of Merge demands more computational complexity and, thus, processing power from the brain, which challenges the stipulation for maximum computational simplicity and energy efficiency, fundamental for the entire minimalist argument.

Notably, although formally stipulated, the notion of simplicity vs. complexity was never articulated with any precision neither in formal terms as simplicity of computation code, nor in terms of algorithmic simplicity as a number of successive operations and ease of implementation, presumably in neuronal activity. Moreover, simple vs. complex are relative notions implying comparison with a measuring standard, articulated in absolute measurable terms. And the stipulation of computational efficiency is difficult to reconcile with the unlimited, or at least very large number of successive operations in terms of brain energy expense.

The argument for atomicity of the LF, i.e. the stipulation that, given its sudden appearance, LF, i.e. Merge and the rules and restrictions on its operations, must have appeared in its entirety at once, with no intermediate stages, is one of the fundamental assumptions in the Great Leap Forward hypothesis, effectively rejecting any evolutionary explanation.

The argument for sudden appearance of the LF is contested by evolutionary biologists who state that complex biological systems take a long time to evolve. A complex entity, which, despite the affirmation of its utter computational simplicity, is in effect a highly sophisticated cognitive equipment, should have taken much longer to evolve, as it involves not only intricate rewiring of the brain, including energy and blood flow redistribution within the brain, but also as a result of these changes, reorganization of the functioning of the entire organism including its nutritional demands, which takes a long time. In other words, time-wise the hypothesis of catastrophic appearance of the language organ is unwarranted.

And even if one accepts the Pinker, Bloom(1990) argument for an evolutionary explanation of the LF by successive mutations, consistent with the Modern Synthesis(MS), an alternative explanation emphasizing that beneficial problem-solving behaviour can be copied by imitation, especially in species with proven learning capacities, is more plausible. Such explanation, articulated by EES, is more convincing, especially given that immediate survival advantages would be afforded by learning instead of genetic inheritance, which is a time-consuming strategy with no immediate survival benefits. In this sense even if a genius mutant demonstrates some novel problem-solving behaviour, assuming the mutation is demonstrated by behaviour (how else would it become available for selection), it is not clear what would distinguish the recursive behaviour of the mutant from learnable behaviours, animal and human. Either way, protracted evolution or genetic accident, the argument for innate UG faces a number of unsurmountable obstacles. Further, learned behaviour can be partially internalized by Baldwin effects, by which fundamental and universal aspects of learned behaviour can be incorporated by genetic, epigenetic, developmental mechanisms.

The argument for recursion as a unique property of language is challenged by a simple observation that recursion is pervasive in nature at large which produces things with nested components e.g. in plant species. It is also pervasive in animal behaviours, e.g. in nest building in birds, in tool making in primates. Recursive, although not unlimited, properties are also easily detectable in human behaviours other than language e.g. musical compositions, visual cognition, the number system, games, social cognition, practically, all domains of human cognition, suggesting that recursion is better defined as a property of general intelligence. Moreover, many social species, e.g. birds, lions etc. live in groups with internal organization with higher and lower ranks, e.g. alpha males and subordinates, suggesting some form of recursive cognition. Primates demonstrate recursive structure in social cognition, e.g. baboons are capable of forming recursive conceptual structures (X is mother of Y (who is mother of Z

(who is mother of me))) (Seyfarth, Cheney, Bergman, 2005). Apes and monkeys have demonstrated deceptive behaviours, suggesting a form of Theory of Mind(ToM)(Baez-Mendoza, Williams, 2020). Primates have demonstrated behaviours which suggest recursive cognition in praxis. They not only use but also invent tools, have mental maps and use these for planning, learn by imitating recursive behaviours, e.g. ,Japanese macaques learn potato washing by observation and imitation(referenced in Ulbaek,1998). Thus, some form of recursive cognition, especially in social species, has deep evolutionary roots and although in humans it has reached unparalleled sophistication, it is not limited to humans and even apes. A plausible explanation for recursive behaviours and linguistic recursion is offered by Arbib (2002) for whom linguistic recursion is a byproduct of increased general intelligence, which allows perception and observation with increasing levels of detail, and makes possible their representation in symbols and discrete hierarchies by linguistic recursion.

“...the NP describing a part of an object may optionally form part of the NP describing the overall object. From this point of view, recursion in language is a corollary of the essentially recursive nature of action and perception once symbolism becomes compositional.” (Arbib, Ibid.).

In this context recursion is explicable with gradual increase in cognitive power and memory space, implying increased general intelligence as evolutionary target.

3.2. Locating LF in the brain

To remind, in biolinguistic circles the standard view of the representation of linguistic knowledge and processing advocates for spatial and functional segregation of language in Broca' s area region of the human brain, identified as the location hosting syntactic computations (Chomsky1968 and elsewhere ; Pinker 1994; Bickerton 2014). The segregationist argument is consistent with the argument for modularity of human cognition (Fodor 1975,1983).

That said, recent empirical studies demonstrate that the localization of language in the brain is difficult to pinpoint given that a large portion of the brain is involved in language-relevant functions, including subcortical regions such as striatum, cerebellum, thalamus, among various others (Fisher, Marcus, 2006). Moreover, the young developing brain is flexible and able to compensate for damaged abilities, linguistic and other wise, as demonstrated by recoveries from injuries, presenting a challenge for the modularity argument. The finding that brain functions are not tailored to the specifics of linguistic syntax, e.g. the brain does not differentiate between pronouns and reflexives (Kaan, 2009), adds to the difficulty. Significantly, the language faculty is said to be “ the most invasive”(Gulyas, 2009, p. 59) i.e. the most widely distributed cognitive faculty in the human brain (Ph. Lieberman, 2000; Bickerton, Szathmari, 2009). Moreover, it is suggested that the widely extended and diversified interconnectivity of the human brain, compared to that of chimpanzees and macaques, makes possible the integration of various types of information (Sherwood et all. 2008). In the same vain Friedericici (2017) made abundantly clear that the human brain is a unified whole of functionally integrated anatomically diverse neuronal assemblages, where perception, emotion, reasoning, socialization, imagination etc. are all interconnected. Moreover, it is demonstrated the representation of language in the brain is well beyond the

physical location and cognitive specialization and integrates multiple processing functions in both hemispheres. Similarly, Lieberman's "unified whole" argument demonstrates that the human natural potential for language is a web of interrelated and spatially distributed groups of neurons, where the same biological entities participate in various neural circuits and support multiple, often unrelated behaviours.

"...a particular neural structure may play a part in different neural circuits that regulate different aspects of behaviour...(and at the same time) distinct neural populations may occur that project to neurons in different brain structures contributing to circuits that each regulate a different aspect of behaviour. " For example, the basal ganglia was found to " switch from one sequence of motor pattern generators to another" (Lieberman 2008, p.218- 219)

And more recent studies confirm that domain-specific mechanisms arise during ontogeny under the influence of experience, forming domain-specific configurations of neuronal connectivity(C. Sherwood et al.2008). So, the natural potential for language, i.e. a language faculty, is a web of neuronal connections with shared functions which have co-evolved in tandem.

From a slightly different angle, the argument for modularity of the grammatical computations or UG, and its direct association with human algorithm for conceptual computation or LoT, in its initial role of organizing the internal life of the human mind, purports isolation from perceptual and emotional experience. Nevertheless, the linguistic computations, as postulated, rely on the pre-existence of the lexicon as a storage of detailed descriptions of real linguistic experiences, which makes it an essential component of the computation procedure. If this is so, the argument for spatial and structural isolation of the language faculty cannot stand as the memory storage does incorporate sensory representations reflecting material reality. In this sense, Port (2007)elaborates that words are stored in memory as rich descriptions of individual examples of perceptual experiences , including phonetic properties of words, where linguistic properties are combined with extralinguistic details. The ingredients of such representations are very concrete, e.g. the idiosyncrasies of the speaker's voice and other sensory representations. The vocal representation of a word in memory is articulation - based , not abstract feature -based. That is, words are stored in memory in the context of specific events with idiosyncratic details. Thus, interaction with reality is inherent to the very functioning of the language algorithm. And the very postulation of the dependence of the syntax algorithm on the lexicon entails that it is by definition intertwined with the rest of cognition. The idea that language system cannot be dissociated from linguistic experiences is formalized in the Parallel Architecture paradigm, which states that " The competence grammar encodes the knowledge involved in the correspondences between phonology, syntax and semantics, and it is the establishment of these correspondences in real time that constitutes the computations that speakers and hearers perform in the course of using language". (Culicover, Jackendoff, 2005 p.163).

In short, the human body and mind have evolved as an integrated unity where language-relevant functions are highly integrated and engage neuronal networks throughout the brain, forming a " language network". Its properties in the adult brain begin their formation early in childhood by initially engaging primarily the right hemisphere (Friederici 2017), most actively involved in perception of prosody and learning basic vocabulary, to gradually integrate

the right hemisphere and Broca's region in processing of complex grammar, only after a decade of experience with language culminating in an adult language faculty demonstrated by adult language proficiency.

At the same time, scholars do not question the prominence of the Broca's and Weernicke's regions in linguistic functions. As per Fedor et al. (2009) although “ the enormous plasticity of the developing brain ...demonstrates that the crucial involvement of Broca's area in syntactical processing in most people cannot be genetically hardwired, rigid condition...It seems more correct to say that some areas of the normally developing human brain are more prone to host and process different components of language than others “ (Fedor et al. 2009. p.300). Thus, the established view that linguistic functions in most normal individuals are asymmetrically concentrated in the left hemisphere is still valid (Fedor, et al. 2009).

3. 3. Reevaluating the saltationist argument

Despite its marginal influence in evolutionary biology, the saltationist argument is given prominence in evolutionary linguistics by some proponents of the biolinguistic perspective, most notably Chomsky, who have argued that the language faculty has emerged at once in its modern form and de novo, marking a complete breakaway from any potential precursors. Consistent with the saltationist argument is the time assessment for its appearance estimated at around 50,000 -100,000 years ago, a short period for biological processes to unfold, effectively precluding any eventual involvement of Darwinian processes. In this context a scenario where a sudden and unprecedented transformational event of major brain reorganization by which a LF miraculously appears in a single individual and is spontaneously propagated in a population, known as the Hopeful Monster hypothesis, is proposed as a plausible explanation. At the same time the LF as postulated by the biolinguistic approach is still a property of considerable sophistication, despite its latest theoretical framing in minimalist terms, (see Hornstein 2018 for summary) which make the Hopeful Monster account highly implausible and which justifies the categorization of such event as supernatural, placing it beyond the realm of scientific explanation(Berwick, Chomsky 2016 and elsewhere).

The saltationist arguments featuring a “hopeful monster” as the progenitor of the human LF was initially conceived as a genetic mutation of transforming consequences, technically consistent with the evolutionary principles of MS. Further, the realization of the limitations of the individual genes in evolution and the futility of the search for a “ grammar gene” in addition to empirical discoveries of the crucial role of development in the formation of the organism, redirected the interest of the generative/ biolinguistic research to the phenotype and its ontogenesis. This resulted in significant theoretical reevaluation of the hypothetical properties of LF. The latest formal paradigm, the Principles and Parameters approach, focuses on the phenotype and identifies two types of language -dedicated cognitive resources, FLN and FLB(Hauser, Chomsky, Fitch 2002 and elsewhere). It injects attention to the developmental aspects of the language faculty (FLN) by postulating a set of genetically determined principles and binary sets of parameters. The universal properties of language systems are attributed to the principles and the observable language diversity is explicable with variation in combinations of parameter settings, determined to occur under experience with a language during the period of ontogenetic development. In this context the gene-centred perspective of the MS is maintained as *Evo-Devo-gen* despite allowing for some environmental influence by turning on a

developmental program upon exposure to a local language(Chomsky 2012 and elsewhere). In the latest revised version of the generative formalism, the Minimalist agenda (Chomsky, 1995 and elsewhere) the role of genes in the formation of the language algorithm is significantly reduced and a major role is attributed to symbol learning, which by its nature is subjected to cultural evolution. An evolution-friendly twist in the saltationist argument is proposed by reassessing the LF as exaptation, i.e. as having initially evolved for some unrelated function and subsequently adapted for grammar. The following possibilities for exaptation are proposed :

- a.** exaptation from conceptual structure, e.g. phrase structure in syntax is believed to be a cognitive exaptation of thematic role structure in protolanguage (agent, theme, goal , etc.) (Calvin, Bickerton, 2000).
- b.** exaptation from the so called social calculus (conceptualization of interpersonal relations) , Dunbar (1998, 2009)
- c.** exaptation from phonology ; Phrase structure , the distinction between Noun Phrase and Verb Phrase is believed to be an exaptation of syllable structure (Carstairs-McCarthy, 1999).
- d.** exaptation of instinctive manual gestures for conscious motor control which suggests that spoken language had gestural origins , emerged initially in the manual domain (in Corballis 2009 , Tomasello 2008).
- e.** Others speculate that the original function of the language organ may have been spacial orientation, i.e.the capacity of tracking is argued to have been put to use in forming linguistic structure.

To note, the Hopeful Monster hypothesis has shown to offer credibility in explaining the origin of flowering plants during the Cambrian period, some 300 mya(Theiben 2006). Nevertheless, the same arguments have questionable credibility for the explanation of the origin of human mind and its highly unusual cognitive and linguistic abilities. This explains why hypothesis for evolution of grammar by a hopeful monster has had limited influence in evolutionary thought.

To remind, as per generative/biolinguistic stipulation the superior cognitive powers of the hypothetical mutant would have been demonstrated by innovative decision-making and problem-solving as behavioural demonstration of higher cognitive abilities, which would justify its survival and preferential reproductive advantages and, in this way explain its preferential perpetuation over generations. That said, one could indulge in speculation that cognitive resources for infinite recursive thoughts, or UG, may have appeared somehow at some point in human evolutionary history. Given that the development and maintenance of brain tissue is the biggest energy expense in the human organism. Moreover, the functioning of a mental capacity of such computational complexity as long strings of recursive thought structures with long-distance dependencies, its development and maintenance, would be extremely energy-consuming, producing an individual with increased energy demands. In this sense, given that the energy supply to the brain is a fixed amount, the organism cannot address the increased energy demands of the mutant brain at a moment notice (as implied by the saltationist argument) as these must come at the expense of vital perceptual and physiological needs, reducing its survival chances. In short, a significant increase of extra energy demand cannot be instantaneously accommodated by the biological body, producing an individual with limited survival potential. And given the assumption of a cognitive capacity for complex reflection, one could imagine a highly intelligent, but socially inadequate loner would not be a

highly desirable mate, reducing its reproductive potential. And reduced survival and reproductive potential would have triggered its elimination by the forces of natural selection. The introvert genius, socially inadequate and physiologically and perceptually deficient is not a good candidate for the founder of humanity.

4. Are formalisms from evolutionary biology readily transferable to evolutionary linguistics?

Ever since Darwin, (in *The origin of species*) scholars have pointed at similarities between the processes of evolutionary change in biological and linguistic contexts, prompting the transplant of conceptual machinery from the theory of evolution into evolutionary linguistics. That said, it is easy to notice that despite indisputable similarities between processes of change in biological organisms and languages, there are also significant dissimilarities (discussed in detail by Comrie 2006) For example :

- a.** The mechanisms of inheritance in linguistic and biological contexts differ significantly. In phylogenesis acquired characteristics are not transmitted to the next generation, while in glossogenesis such transmission is the norm, since linguistic innovations of predecessors are perpetuated by the new generation. Thus, biological evolution is Darwinian-Mendelian, glossogenesis is Lamarckian.
 - b.** In phylogenesis the genomes of the parents are inherited in their entirety at the single point of conception, while linguistic items are learned piece-meal at different points of the learning process.
 - c.** Linguistic entities are transmitted (inherited) via language use, that is, mediated by communicative interaction as conscious behaviour, while genes are inherited directly, by instinctive urge to produce offsprings.
 - d.** In genetic transmission the offspring has no active role in the production of the new genotype, while in the formation of the idiolect the learner is active participant by making choices in reflection of one's individuality.
 - e.** In life forms an organism inherits characteristics of two parents, while in the formation of the idiolect multiple community members, e.g. parents, extended family, neighbours, teachers etc. make linguistic contributions. Moreover, in biological organisms the offspring inherits the totality of the parents' genomes, while in the idiolect the contribution of the parents' idiolects is only one of many.
 - f.** In life forms genetic variation is produced by recombination of the pool of parental genes, while in the formation of the idiolect the learner can introduce his/her own innovations, although these are restricted by what would be considered acceptable by the community. The formation of the idiolect, then, is a creative process and a reflection of one's individuality as a creative entity.
 - g.** In phylogenesis copying errors are rare and the genome is highly conservative, while languages display a much higher intra-lingual diversity of idiolects and sociolects which amplifies the potential choices for selection.
 - h.** The selection factors in phylogenesis and glossogenesis are very different. Although biological factors play a role in both processes, in glossogenesis the social and cultural environment is of primary importance.
 - i.** The number of species of flora and fauna is in the millions, while diversity of languages is
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highly constrained at about 7000.

In sum, despite similarities the evolution of language cannot be reduced to biological processes, i.e. the theoretical instruments of evolutionary biology are insufficient for understanding the mechanisms of language evolution. Bottomline, although metaphorical transplanting of theoretical paradigms between research fields is a common practice, such methodological decisions are not always wise.

5. The founder effect and the evolution of language diversity: the hypothesis of a “mother tongue” and its challenges

Some students of language evolution have adopted the population genetics paradigm in attempt to trace the origin and diversification of language systems, prompting the hypothesis for monogenesis, i.e. today's diversity of languages originated from a “mother tongue”. This led to the hypothesis that all modern languages have branched out of the so called protoworld, or a “mother tongue”, the language of an ancestral founder population, whose traces, or linguistic genes, are said to be preserved in modern languages, proposed by Merritt Ruhlen (1994). It is not taken seriously by most linguists, given the diversity of local environments and thus, diversity of adaptive responses by communities, including adaptation of their languages in both lexicon and grammar, not to neglect the fast pace of language change, which makes preserving linguistic forms for some 100, 000 years highly unlikely.

Nevertheless, despite observable diversity, common traits at all levels of languages, phonology, semantics, grammar are found. For example semantic universals pinpointed by Wierzbicka (1992, Goddard (2007) are explicable with the fact that we all inhabit the same planet and our minds reflect nature on earth with anthropological bias, in reflection of the basics of human existence which results in constructing universals of human semiosis. Similarity in phonetic and phonological aspects of languages, e.g. the basic CV syllable structure is a universal, explicable with the innate properties of the human speech capacities. Similarity in (Jablonka, Dor 2000) some near-universal grammatical concepts are explicable with the similarities of human minds to generalize and form abstractions and universal principles of historical changes in grammars are explicable with common processes of grammaticalization (Heine, Kuteva 2007).

That said, this does not necessarily suggest that these are likely the remnants of a “mother tongue” invented by a single founding population, but instead, given the commonalities in language ecologies (language ecology in Mufwene's sense) such linguistic near universals were inevitable to have emerged independently within multiple communities. A strong indication of such likelihood is the diversity of the newly invented sign languages, observable today. Thus, the dilemma on monogenic vs. polygenetic account of language origins could, on the basis of information currently available, as pointed out by Mufwene, (2014) “does not suggest that modern humans emerged in one single specific location in East Africa” and unless or until new discoveries point to the contrary, the argument for polygenic origins of humanity and language is the stronger.

In a similar vein Atkinson (2011) finds parallels in the evolution of genetic and linguistic diversity within populations after the dispersal of humans out of Africa and argues that patterns in diversification of populations as language speakers coincide with patterns of diversification of language systems. For example the migration out of Africa led to establishment of multiple

populations with reduced membership, which coincides with reduction of diversity, genetic and linguistic, most notable in reduced phonemic inventories. In this sense multiple repetitions of "a cultural founder effect operated during our colonization of the globe" (ibid. p. 348) explains the social, cultural and linguistic diversity we witness today.

5.1. Languages as populations

The field of historical linguistics readily adopts conceptual instruments from evolutionary biology for understanding periodical changes in language systems, although different interpretations of alignment between biological and linguistic concepts is proposed by different authors. Croft (2000) applies evolutionary principles in the study of glossogenesis where grammar is defined as analogue to the genome, and language, in terms of utterances, analogue of the phenotype. Croft introduces a population perspective, although his model deviates from the original population genetics model, adapted to fit the context of language and its primary role as communication system. Here language is defined as a pool of linguemes, analogous to a population, where linguistic forms of various types exist and replicate in coordination as parts of a population of linguemes. Evolving entities are defined not by the nature of *replicators* (analogue to genes) but by the nature of their *interactors* (analogue to phenotypes) and their interactions. In this sense the replicators are nested in structures of ever growing complexity such as phonology, syntax, morphology. The replication of a phoneme requires coordinated replication of other phonemes and the replication of words, similarly, requires the coordinated replication of other such members of the lexicon and the replication of lexical words, which form part of an idiom, is conditioned on the replication of the idiom. There is mutual dependence of phonemes and morphemes as morphonological rules.

From evolutionary perspective Croft's model of glossogenesis/language change adopts the original Darwinian theory by defining language change in terms of variation, competition and inheritance. **a.** variation in various levels of language use is pervasive: 1. there is diversity of idiolects, reflecting individuality of characters, 2. diversity of sociolects, motivated by various social factors: age, gender, class, 3. variation within idiolects: in phonetics, lexicon and grammar as the same person uses different linguistic instruments at different occasions in expressing the same idea. **b.** competition: two or more vocabulary items are used as alternatives to encode the same concept (synonyms), or various syntactic structures are alternatively used to describe the same event. **c.** selective inheritance: some linguistic forms are persistently passed on over generations, which makes possible the understanding between generations, and texts written centuries ago and allow historical linguists to determine language families.

5. 2. Languages as species within a language ecology

An analogy between species and languages from a perspective of niche construction is proposed by Mufwene (2005, 2014 and elsewhere) who defines a language as a population of linguistic entities residing in the minds of speakers. The term "language ecology" is introduced, reminiscent of the concept of ecological niche, i.e. the context, e.g. communicators' bodies and minds, population numbers and structure, economy, local climate, communicative pressures, in which languages are embedded and with which they co-adapt.

Among these, “the human anatomy and mind are the most fundamental ecological factors ...as they filter...all other external ecological factors that bear on language evolution” (2014, p.19). The appearance , proliferation and disappearance of languages depends on the life and death of the speakers as hosts of linguistic forms, although the death of a language does not necessarily result from biological extinction as many examples exist when a community stops using the language of its ancestors and adopts a different one, e.g. if it confers a higher social and/or economic status. In this sense a significant contribution to evolutionary linguistics is the understanding of language in its bio-cognitive and socio-cultural aspects and language evolution as integration of biological and cultural processes.

In sum, adopting theoretical machinery from fields of inquiry as distant as biology into linguistics needs careful tuning and modification.

6. Connecting the family trees of populations and languages

Scholars today tacitly assume close correlation of genetic and linguistic family trees, i.e. that the genetic diversity closely resamples linguistic diversity, which has justified the metaphorical use of the Darwinian theory by evolutionary linguistics, where a crucial role is attributed to the environment as selective factor. That said, an anti-adaptationist account is preferred by population genetics where a “ neutral model” redefines evolution within the genetic constraints of a small population by extending the propagation of allele variants with higher frequency, i.e. a genetic drift, neutral or irrelevant to environmental adaptability. Within evolutionary linguistics the “neutral model” is argued to parallel iterated language learning, where regularities in language systems emerge by extending the distribution of linguistic variants with higher frequency of use by the new generation of language learners from examples provided by the previous generation. In this sense the processes of biological evolution as altering the genetic makeup of a population by extending gene frequencies, and cultural evolution of languages as extending patterns of grammaticalization by language learning, are shown to display highly similar dynamics(see Reali , Griffiths 2010). Tracing genetic histories of populations are addressed by cross-inferences with linguistic markers and vice versa. Empirical justification for such theoretical stipulations have confirmed a broad pattern of overlap, i.e. “ global phylogeny of human populations showed some broad matches with a global language tree”(Barbieri et al. 2022, p.1) (first argued by Cavalli-Sforza, Menozzi, Mountain (1988), given that as a general rule individuals pass on to their offsprings their genes as well as their cultural values and languages. Notably, distribution of genes and languages also has a geographic aspect as genetically related populations usually live in spatial proximity. Calculations of genetic and territorial distances of populations post migration out of Africa are offered by Cavalli-Sforza(1997).

Convergence and coordination of biological and cultural evolution is traced from lower branches of human evolutionary tree, starting from the diversion from primates as bio-cognitive adaptation to novel environments , e.g. restructuring of the skeletal architecture in adaptation to bipedalism, especially enlargement of the cranial cavity to accommodate continuous brain enlargement. A consequent higher demands for energy is correlated with adoption of new food sources, affording more complex cognitive performance, demonstrated by developing new tools, techniques for better subsistence. These evolve in coordination with socio-cultural innovations, e.g. invention of weapons, adornments. Increased cognitive abilities and

intensified interactions spurred more detailed semiotic representation of reality, natural and social, driving the formation of symbols from motivated modes of representation, e.g. iconic, indexical signs. This, coupled with increased vocal dexterity, made possible larger phonetic arsenal and reflected in enlarged vocabulary, which became inevitably reflected in the shape of the language systems, demanding introduction of more structured linguistic communication, e.g. grammaticalization. The rise of some cultural sophistication produced cultural rituals, myths, stories, art.

The consensus is that humans are descendants of a small population inhabiting Africa some 100-200kya(thousand years ago). It is estimated that the Khoisan peoples, who are the current inhabitants of the same territory, are the direct genetic inheritors of the human founder population, and for the proponents of the founder effect to be a rich source of information for determining the genetic makeup of this population. Moreover, under the assumption that the Khoisan tribes, speakers of click languages, have preserved the cultural heritage of the ancestors, including linguistic heritage, and given the general pattern of overlap of genetic and linguistic histories, for the proponents of monogenesis of language diversity, one is inclined to suppose that **a.** the language of the Khoisan peoples could contain remnants of the proverbial “mother tongue” and **b.** the “mother tongue” was likely a click language.

That said, migration and resulting from it territorial distancing inevitably brings diversification of populations and languages, obscuring a straight line of inheritance, especially pronounced in languages. In addition, a pattern of decreasing genetic diversity with increasing distance from Africa is noticed, attributed to repeated genetic drift and genetic isolation (Reyes-Centeno et al. 2016). And isolation is shown to affect genetic and linguistic trees differently, e.g. “There are fewer (language) families in the linguistic tree than there are populations in the genetic tree” (Cavalli-Sforza 1997, p.7721), e.g. “Europe has a genetic variation in general about three times lower than that of other continents” (see more examples in ibid.p.7721). This is suggestive of **a.** divergence in the regularity in patterns of genetic diversification, due in some occasions to conquest and increase in population growth of the occupier over the occupied populations, resulting in disproportionate proliferation and dominance in the gene pool. In other cases isolation between two populations is incomplete and continuous migration leads to continuous gene flow from neighbouring populations, e.g. the genetic makeup of the population of the USA is a mixture resulting from centuries of gene flow from African slaves and European settlers. **b.** Divergence in the correlation between genetic and linguistic diversity is due to the fact that populations and languages are often affected differently by historical and cultural factors, e.g. population replacement is rare while language replacement happens often when a culturally and economically dominant population imposes its language over the dominated, e.g. by conquest and/or by higher social or economic status. As an example the languages of native populations in Australia and America are almost extinct under the dominance of English, Spanish, Portuguese, languages of the colonizers. Thus, the regularity of mutation rates, assumed to apply to all circumstances, is disturbed by the processes unfolding in individual circumstances at population level. Additional examples are provided by “studies (which) have repeatedly identified instances where languages and genes clearly do not match”(Barbieri et al. 2022, p.1), e.g. Hungarian language belongs to a family of Siberian languages while its speakers in central Europe trace their genetic ancestry to European populations.

Culturally driven language shift is not uncommon, given that a new language is learned within a generation. This suggests that the histories of biological species and human populations are different processes which follow different patterns and divergence of these two types of patterns is to be expected. These findings are significant enough to justify the conclusion that “the association between genes and languages is the exception rather than the rule” (Barbieri et al. 2022, p.2).

At the same time others demonstrate partial convergence of patterns of change in both populations and languages. For example Reyes-Centeno et al. (2016) analyze the effects of population's evolutionary history in the shape of the human skeleton and finds that genetic changes are reflected in the shape of the skull, while the facial structure is less influenced, largely neutral, to genetic and more dependent on environmental factors. And facial structure is found to be correlated with language, especially with the lexicon, where language-face alignment is interpreted as largely unaffected by genetic evolution and under a significant influence of environmental selection. So, different patterns of association of genes and languages are detected for different parts of the phenotype.

In addition, population size was found to be a factor in both contexts, e.g. in larger populations basic vocabulary expands rapidly while in small populations basic vocabulary contracts rapidly. This pattern coinciding with patterns in biological evolution predicts a larger genetic diversity in large populations and reduction of genetic diversity in smaller populations, produced by genetic isolation, a pattern overlap demonstrated in related populations and languages on various neighbouring islands of Polynesia (Bromham et al. 2015)

To remind, the population-genetic approach stipulates that isolation produces evolutionary changes driven by internal dynamics in both genetic makeup of the populations and languages (internal in the sense isolated from the broad environment), given that populations inhabiting the same or very similar environments often follow different genetic and linguistic trajectories. At the same time although biological as well as cultural innovations are said to propagate unaffected by selection, argued by anti-adaptationist accounts, the survival and longevity of such novel biological and cultural creations must still be filtered through selection and demonstrate superior survivability to competing alternatives in the respective environments, e.g. natural habitats and human minds.

In sum, very similar processes operating on different ontological entities produce different outcomes.

7. The complexity theory, a challenge to the founder effect

Some usage-based perspectives within evolutionary linguists adopt the founder effect hypothesis for understanding the formation of language diversity as diversification from a pattern of communicative conventions from the cultural heritage of a single founder population, or top to bottom. As an alternative the formation of sociolects is explained within the complexity theory, initially designed as a theoretical framework charting the transformations in physical matter from a state of chaos or disorder into internally organized complex unities with multiple integrated components acting in coordination. Change is here understood in terms of series of events of self-organization, or structure formation, resulting from the internal dynamics in a closed group of interacting entities. The theoretical framework of the complexity theory was recruited for explaining cultural phenomena as spontaneous

emergence of cultural conventions. Heylighen (2013) has designed a model where cultural conventions emerge by self-organization from interaction of intentional agents as coordinated efforts of individuals aligned on a unifying goal. The model is meant to show how social and linguistic conventions are formed and cultural differences emerge and shape languages and cultural identities of the communities. Cultural self-organization happens as all interacting individuals contribute to the final result, thus, no one entity or individual imposes its control. In addition, small changes in cultural and linguistic behaviours, which initially begin from local interactions, or bottom-up, can snowball and produce disproportionately large effect on the final result, which makes them inherently unpredictable. Similarly, Keller (1994) presents the “invisible hand” hypothesis where a sociolect is defined as a rule system emerging spontaneously as unintended byproduct of the rational and intentional interactions of individual speakers to communicate with fellow community members in an attempt to solve interpersonal problems at a local level. In sum, here languages are understood as products of communities not of individuals.

Summary and conclusions

For most scholars evolution by Darwinian principles of gradual accumulation of minor adaptive improvements is accepted as the standard explanation for the observable diversity of life forms. That said, the explanation of the notable gap between the cognitive abilities of us and our evolutionary predecessors and especially the unparalleled human linguistic abilities by evoking Darwinian gradualism has been unconvincing to some which is why an alternative explanation involving a founder effect, initiated by a single individual, a “hopeful monster” has been proposed. The saltationist argument has been particularly influential among some linguists within the generative/ biolinguistic perspective in attempts to explain the unprecedented human capacities for language.

That said, the hypotheses proposed by the biolinguistic formalism have been challenged by empirical demonstrations which show divergence from the formal stipulations and demonstrate that there was never a single evolutionary event transforming pre-human into human species, but series of small and cumulative changes, resulting from multiple events of demographic isolation, or founder effects in the course of human evolution during Pleistocene. Moreover, human speciation is found to be a result of a combination of a number of evolutionary processes. A role for hybridization as a potential source of variation, and genetic drift, found to have been major factors in the formation of the human phenotype, predominantly with neutral selective consequences, in interplay with selection offers a novel, integrative approach, presenting a significant challenge for the founder effect as a credible explanation of human evolution.

Such challenge is extended to the bio-linguistic argument as explanation for the human capacity for language, i.e. both Language of Thought and Universal Grammar.

At the same time the interpretation of the founder effect for explaining the diversification of languages from a “mother tongue” has challenges of its own given the demonstrable divergence of patterns of diversification in human populations and languages. In this context language diversity is better explicable with emergence of social conventions by self-organization.

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